Advanced Labs I (AY2025)

Theme			
X-ray diffraction			
Dates of Experiments	October 27, November 7 and 17 2025	Report Submission Date	24/11/2025
Student ID Number : 202318030		Name : Andrea Trevanna	

Co-experimenters	
Student ID Number	Name
202318030	Nguyen Nguyen Cong
202118028	Woo Been Cha

Date of Receipt	
Recipient	

By the theme instructor,

Date of return	
Rating /10	
Are the principles, equipment, and experimental procedures fully understood? a b c d e	
Are the experimental data properly analyzed? a b c d e	
Are the experimental results presented appropriately? a b c d e	
Is the discussion substantial? a b c d e	
Is the structure of objectives, methods, results, discussion, etc. appropriate? a b c d e	
Is it carefully described? a b c d e	
Comments	

Advanced Labs 1 X- Ray Diffraction Report

Objective: Acquire the basic techniques of X-ray diffraction experiments and obtain diffraction phenomena and the fundamentals of crystallography. Two methods are done in this experiment, Laue Method and Diffractometer method.

A. Experimental Contents and Methods

1) Laue method: Laue photographs of LiF and Si single crystals are taken
2) Diffractometer method: First a standard sample of silicon is measured. The observed diffraction pattern is indexed and the lattice constant of Si is derived. Next, two "unknown" samples are measured. By analyzing the diffraction patterns, the substances and their crystal structures are identified.

    1) Diffractometer method: The diffraction of monochromatic X-rays by powdered specimen is measured.
    2) Laue method: The diffracted beams form an array of spots on the film, called Laue pattern

Method	Types of X-rays	wavelength λ	Sample	Detector	2θ
Laue	white (continuous)	Variable	single crystal	film or camera	fixed
Diffractometer	monochromatic (characteristic)	fixed	powder (polycrystalline)	counter	variable

B. Week I: Laue Method

In the first week we focused on the Laue Method to identify the indices of Laue spots generated by the crystals, namely LiF(001), Si(001) and Si(111). Then, we analyzed the symmetry present with these Laue patterns. The Laue Photo was generated by firing an X-ray beam directly perpendicular to a thin flat piece of the crystal. We identified the symmetry of the Laue patterns if any was present by comparing the Laue spots on the photo to the poles

present on the standard projection of a cubic crystal of miller index (001). We only calculated the miller indices for the LiF sample because Si had a different crystalline structure to LiF and we only had a (001) standard projection.

A) Instruments and Methods

The three samples that are used for the Laue method are flat plate Silicon (001) flat plate Silicon (111), and flat plate Lithium Fluoride (001).

First, we used an X-ray beam, using the X-ray machine shown in figure 1, that was directed onto a perpendicular thin crystal of LiF (001. The X-ray is produced by a cathode-filament tube. which produces a white, continuous X-ray with a variable wavelength λ. The crystal diffracted the X-ray onto a CMOS camera 3cm away which we used to take a Laue photo. We analyzed the photo to identify Miller indices and symmetry.

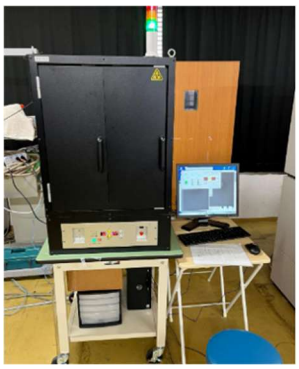


Figure 1 X-ray machine used for Week 1

We set the voltage value to 30kV, current to 20mA, exposure time to 5 seconds and the number of shots to 3 using the control software. We repeated this for the samples of Si (001) and Si (111).
The information obtained from this experimental method is a Laue Photo that describes the structure of the lattice (including symmetry, shape and orientation) that was analyzed with a stereographic projection using a ruler.

B) Experimental details

Measurement of (001) and (111) planes of Si and (001) plane of LiF is done as follows

1. Fix the sample with compound. The sample is a thin flat plate less than 1 mm thick, with a face corresponding exactly to the (001) or 111) plane. To orient the sample so that the X-ray beam is directed perpendicularly to the plane. Aligning the inclination of the sample first by eye, then by fine-tuning the angle of goniometer using a dedicated screwdriver.

2. Ensure that the distance between the sample and the X-ray CMOS camera is 3 cm. 3. Ensure that the beam stopper is attached to the camera to prevent direct beam to the camera, and close the door.
4. Set the voltage and current to the values 30kV and 20mA, respectively.
5. Set the exposure time (5 seconds) and the number of shots (3 shots) using the software that controls the camera.
6. Set the time (30 seconds) for the X-ray shutter to open using the control software of the X-ray generator, and open the shutter.
7. While the shutter is open, take Laue photos using the camera's control software.
8. After the photo is taken, save the photo.
9. Change the sample and repeat from step 1 for all 3 samples.

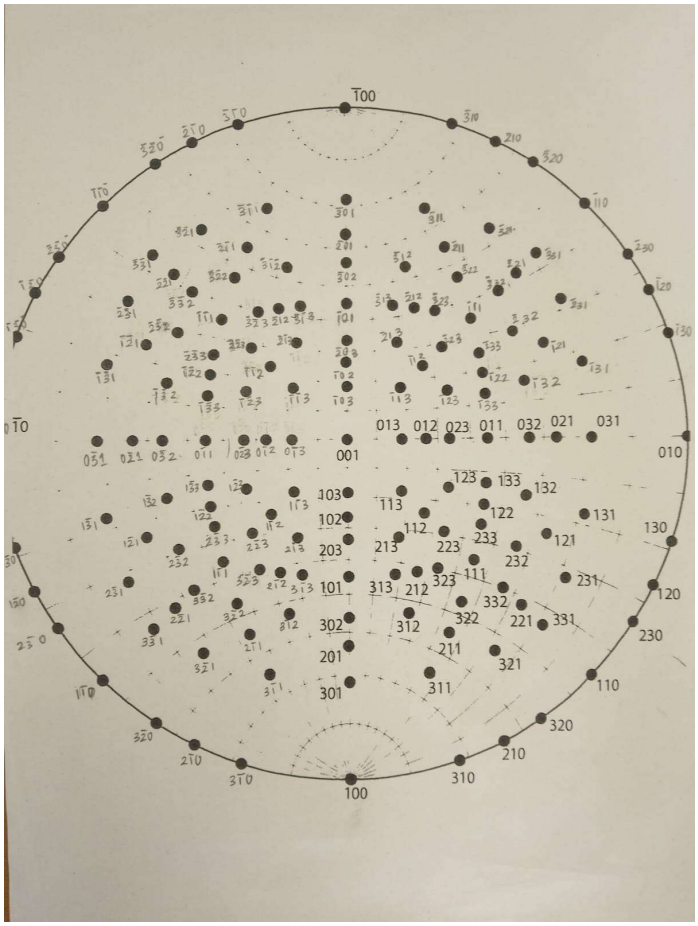


Figure 2 Standard preojection

For the indice numbering of LiF, the following standard projection (Figure 2) and the stereographic ruler (Figure 3)is used.

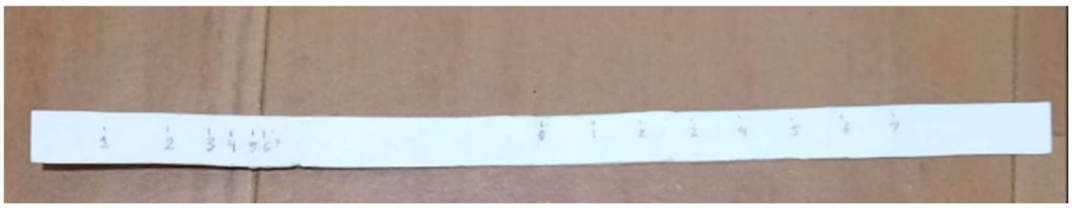


Figure 3 Stereographic ruler

C. Week 2: Diffractometer method

In the second week, we used the diffractor method to identify various crystals by changing the angle at which the X-ray hits the crystal tested and measuring the corresponding intensity of the reflected X-ray. Using the results and software

(PANalytical High Score),we get a Diffraction Pattern which is a graph of different intensity peaks. To identify each crystal type, the angle of each intensity peak is used. Using the results we derive the indexing patterns using the formula:

$$2d_{hkl} \sin\theta = \lambda$$

and the derivation of the lattice constant a using:

$$\frac{\sin\theta^2}{h^2+k^2+l^2} = \frac{\lambda^2}{4a^2} = const$$

We start the experiment with a known sample of Si-crystal powder, and then move on to two unknown samples of Cu and Mo.

A) Instruments and Methods

In week 2, we measured the diffraction pattern of a sintered Si plate for θ range

between 20◦ and 150◦. We used the machine shown in Figures 4 and 5(inside of the machine).



Figure 4 Week 2 X-ray machine

This machine produces a monochromatic, characteristic X-ray with a constant wavelength. The x-ray passes through soller slits to make sure that the x-rays are

parallel to each other. It also passes through a Ni filter to monochromize the radiation. We set the power to 40kV and the current to 10mA. The resulting diffraction intensity is detected by a counter. We used specialized software to identify peaks and form them into a list . Then we analyzed the results using Excel and found the Miller indices and structure factor which will be discussed in detail later. We repeated the experiment for two unknown crystals.

B) Experimental details

Experimental procedure for the difractometer method is as follows:

1) Measurement of difraction patterns

• The standard sample: Measure the diffraction pattern of a sintered Si plate for 29 range between 20° and 150°. The power of X-ray beam should be 40kV, IDenA.

. • Unknown samples: By following the same procedure, measure the diffraction patterns of powdered samples of Cu and Mo as unknown samples.

2) Indexing patterns : The angle of each peak in diffraction pattern is related to the inter-planar sparing dhkl by the Bragg's law: $2d_{hkl}\ sin\theta = 1$. Assign the appropriate triple (hkl) of Miller indices of the lattice plane to each of the experimental diffraction peaks.

3) Derivation of lattice constant Calculate $sin^2\theta$ for each diffraction peaks; this set of $sin^2\theta$ values is the ram material for the determination of the lattice parameter a. The observed values of $sin^2\theta$ always contain small systematic errors, and the values of a calculated for the various peaks on the diffraction pattern approach the true value more closely as 2θ increases. The most accurate value of a can be found by plotting measured values against certain function of θ, the extrapolation function, and extrapolating it to 2 θ = 180º

D. Results and Analysis

1) Laue Method

i) LiF (001)

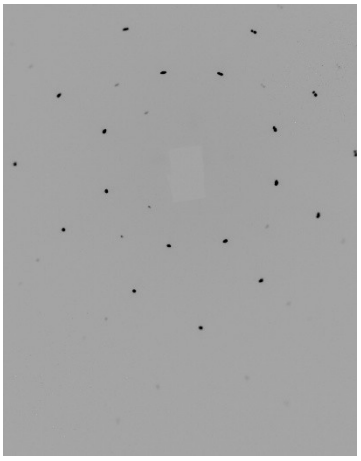


Figure 5 Laue photo of LiF (001) projection

The actual projection image looks like Figure 5. Lithium Fluoride (001) has a 4-fold symmetry which can be observed in the above image. To clearly showcase this symmetry and figure out the precise crystal orientation, the Laue photograph was annotated using a stereographic ruler with reference to a standard projection (Figure 2) by placing it on top of a tracing light box. However, to obtain a better symmetry, we had to repeat and do another Laue photo of LiF. The annotated image can be seen in Figure 6.

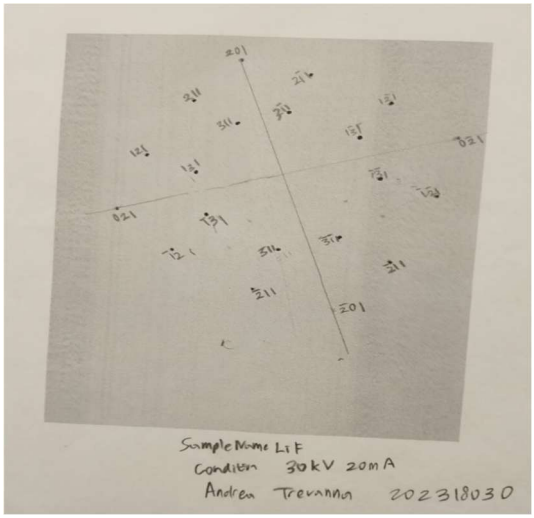


Figure 6 Annotated crystal orientation and symmetry of LiF (001)

ii) Silicon (001)

Silicon (001) has a similar structure to LiF (001) albeit more complex. It has a 4-fold symmetry which can be observed in the Laue photograph in Figure 7. The dotted lines represent a digital annotation to better showcase the 4-fold symmetry and was not part of the original output.

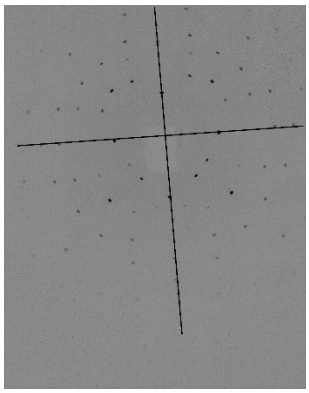


Figure 7 Silicon 001 projection and Symmetry

iii) Silicon (111)

Silicon (111) has a 3-fold symmetry as in Figure 8. As in the Laue photo for Silicon(001), straight lines are added to indicate the symmetry.

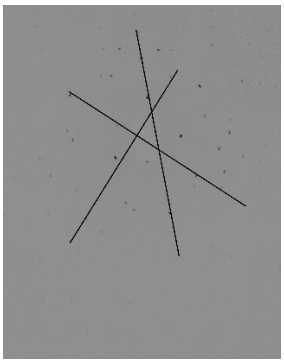


Figure 8 Silicon 111 projection

We only index the Laue spots for LiF(001) because we only have a 001 projection and the structure of Si is too complicated.

2) Diffractometer method

The data for the three samples is collected using the PANalytical software and analyzed using X'Pert High Score software to recognize diffraction peaks. The Excel data is recorded and analysed for Si and the two unknown samples (later determined as Cu and Mo).

i) Silicon

The diffraction pattern and peaks of Si detected and analysed by High Score software is in the figure below (Figure 9). For the peak detection, we also applied linear fitting.

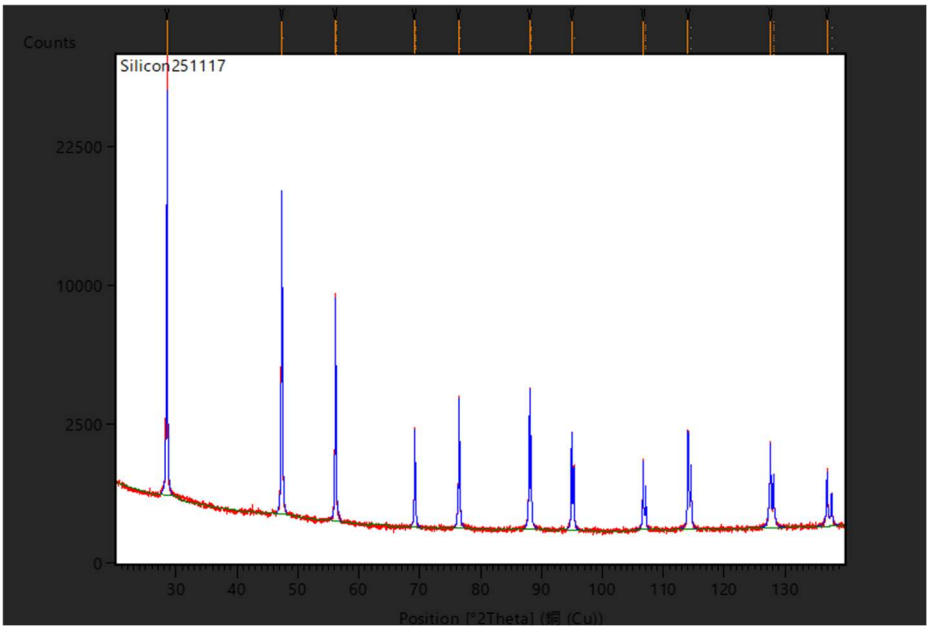


Figure 9 Diffraction pattern and peaks detection of Si

The diffraction data from High Score is then copied and extracted into Excel (Table 1). This includes the degrees of position which would be used for analysis.

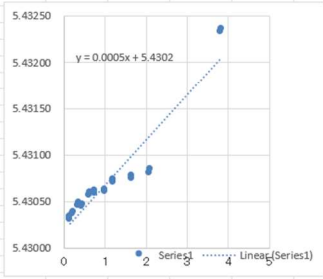
Table 1data extracted from High Score

No.	Pos. [° 2T	FWHM Lef	計算面積	指定	h	k	l	多重度	F 観測値	F 計算値	Area[cts*° 2T	由来	バックグラウンド[cts	d値 [Å]	NET強度	K-A2/K-メー一致		一致条件	相対強度	データ元	ステータス	積分幅 [° 2T
1	28.4331	0.0671	3040.99						0	0	3040.99	理論KA1	602.78	3.13655	25928.6	0.5	FALSE		100	プロファイルフィ使用		0.11728
2	28.5053	0.0671	3040.99						0	0	1520.5	理論KA2	602.78	3.12878	12964.3	0.5	FALSE		50	プロファイルフィ使用		0.11728
3	47.3008	0.0752	1862.33						0	0	1862.33	理論KA1	313.93	1.9202	17271.4	0.5	FALSE		66.61	プロファイルフィ使用		0.10783
4	47.4256	0.0752	1862.33						0	0	931.16	理論KA2	313.93	1.91544	8635.71	0.5	FALSE		33.31	プロファイルフィ使用		0.10783
5	56.1207	0.0816	1059.62						0	0	1059.62	理論KA1	231.22	1.63754	8924.71	0.5	FALSE		34.42	プロファイルフィ使用		0.11873
6	56.2726	0.0816	1059.62						0	0	529.81	理論KA2	231.22	1.63348	4462.36	0.5	FALSE		17.21	プロファイルフィ使用		0.11873
7	69.1284	0.0914	284.29						0	0	284.29	理論KA1	166.65	1.35776	2140.47	0.5	FALSE		8.26	プロファイルフィ使用		0.13282
8	69.3247	0.0914	284.29						0	0	142.15	理論KA2	166.65	1.3544	1070.24	0.5	FALSE		4.13	プロファイルフィ使用		0.13282
9	76.3762	0.0877	444.79						0	0	444.79	理論KA1	161.02	1.24594	3392.89	0.5	FALSE		13.09	プロファイルフィ使用		0.13109
10	76.6003	0.0877	444.79						0	0	222.39	理論KA2	161.02	1.24286	1696.44	0.5	FALSE		6.54	プロファイルフィ使用		0.13109
11	88.0305	0.0999	579.32						0	0	579.32	理論KA1	154.99	1.10858	3816.13	0.5	FALSE		14.72	プロファイルフィ使用		0.15181
12	88.3059	0.0999	579.32						0	0	289.66	理論KA2	154.99	1.10584	1908.07	0.5	FALSE		7.36	プロファイルフィ使用		0.15181
13	94.9533	0.1082	346.67						0	0	346.67	理論KA1	142.87	1.04518	2069.84	0.5	FALSE		7.98	プロファイルフィ使用		0.16749
14	95.2642	0.1082	346.67						0	0	173.34	理論KA2	142.87	1.04259	1034.92	0.5	FALSE		3.99	プロファイルフィ使用		0.16749
15	106.713	0.1239	228.04						0	0	228.04	理論KA1	143.97	0.96004	1198.93	0.5	FALSE		4.62	プロファイルフィ使用		0.19021
16	107.097	0.1239	228.04						0	0	114.02	理論KA2	143.97	0.95766	599.47	0.5	FALSE		2.31	プロファイルフィ使用		0.19021
17	114.097	0.1316	450.8						0	0	450.8	理論KA1	144.8	0.91797	2128.53	0.5	FALSE		8.21	プロファイルフィ使用		0.21179
18	114.538	0.1316	450.8						0	0	225.4	理論KA2	144.8	0.9157	1064.26	0.5	FALSE		4.1	プロファイルフィ使用		0.21179
19	127.555	0.1817	478.97						0	0	478.97	理論KA1	163.81	0.85867	1666.41	0.5	FALSE		6.43	プロファイルフィ使用		0.28743
20	128.136	0.1817	478.97						0	0	239.49	理論KA2	163.81	0.85654	833.21	0.5	FALSE		3.21	プロファイルフィ使用		0.28743
21	136.909	0.1978	269.77						0	0	269.77	理論KA1	179.84	0.82817	881.05	0.5	FALSE		3.4	プロファイルフィ使用		0.30619
22	137.636	0.1978	269.77						0	0	134.89	理論KA2	179.84	0.82611	440.52	0.5	FALSE		1.7	プロファイルフィ使用		0.30619

The first column from Table 1 is used for the 2θ row in a new table in Excel which is used for the analysis of Si diffraction and determination of the lattice parameter a, which is specific to each element (Table 2). 2θ represents twice the angle of incidence at which X-ray peaks occur.

Table 2 Si diffraction analysis table in Excel

λ(Å)	2θ	θ	radian	$\sin^2\theta$	$\frac{\sin^2\theta}{\sin^2\theta_1}$	trialS1	$\frac{\sin^2\theta}{\sin^2\theta_1}s_1$	Integer s	hkl	$\frac{\lambda^2}{4a^2}$	a(Å)	Extrapolate	d_{hkl}(Å)	observed	reported	error(%)
1.54051	28.4331	14.2166	0.24813	0.06031	1	3	3	3	111	0.0201	5.43237	3.82628	3.1364	5.4302	5.4282	0.03683
1.54433	28.5053	14.2527	0.24876	0.06061	1.00498		3.014939163	3	111	0.0202	5.43233	3.81557	3.1364			
1.54051	47.3008	23.6504	0.41278	0.16093	2.66816		8.004485118	8	220	0.02012	5.43084	2.09165	1.9201			
1.54433	47.4256	23.7128	0.41387	0.16173	2.68145		8.044337051	8	220	0.02022	5.43081	2.08447	1.9201			
1.54051	56.1207	28.0604	0.48975	0.22128	3.66881		11.00642155	11	311	0.02012	5.43078	1.65544	1.6374			
1.54433	56.2726	28.1363	0.49107	0.22238	3.68707		11.06121012	11	311	0.02022	5.43075	1.649	1.6374			
1.54051	69.1284	34.5642	0.60326	0.32186	5.33652		16.00956518	16	400	0.02012	5.43074	1.19532	1.3577			
1.54433	69.3247	34.6624	0.60497	0.32346	5.36308		16.08923311	16	400	0.02022	5.43071	1.18954	1.3577			
1.54051	76.3762	38.1881	0.66651	0.38223	6.33737		19.01211977	19	331	0.02012	5.43063	0.99924	1.2459			
1.54433	76.6003	38.3002	0.66846	0.38413	6.3689		19.1067017	19	331	0.02022	5.43061	0.99369	1.2459			
1.54051	88.0305	44.0153	0.76821	0.48282	8.00515		24.01546324	24	422	0.02012	5.43062	0.74431	1.1085			
1.54433	88.3059	44.153	0.77061	0.48522	8.04498		24.134944	24	422	0.02022	5.43059	0.73902	1.1085			
1.54051	94.9533	47.4767	0.82862	0.54317	9.00586		27.01757456	27	333	0.02012	5.43060	0.61985	1.0451			
1.54433	95.2642	47.6321	0.83134	0.54587	9.05066		27.15198963	27	333	0.02022	5.43057	0.61465	1.0451			
1.54051	106.713	53.3565	0.93125	0.64379	10.6741		32.02230443	32	404	0.02012	5.43047	0.44395	0.9600			
1.54433	107.097	53.5483	0.93459	0.64699	10.7272		32.18161717	32	404	0.02022	5.43045	0.43887	0.9600			
1.54051	114.097	57.0485	0.99568	0.70414	11.6747		35.02421673	35	531	0.02012	5.43049	0.35258	0.9179			
1.54433	114.538	57.2688	0.99953	0.70764	11.7328		35.19849834	35	630	0.02022	5.43046	0.34754	0.9179			
1.54051	127.555	63.7774	1.11313	0.80476	13.343		40.02906563	40	602	0.02012	5.43039	0.21764	0.8586			
1.54433	128.136	64.0679	1.1182	0.80876	13.4094		40.22821387	40	602	0.02022	5.43037	0.21265	0.8586			
1.54051	136.909	68.4544	1.19475	0.86513	14.344		43.03204332	43	533	0.02012	5.43034	0.145	0.8281			
1.54433	137.636	68.8179	1.2011	0.86944	14.4154		43.24615885	43	533	0.02022	5.43032	0.14002	0.8281			



From 2 θ , the rest of the data can be calculated. The two wavelengths used are λ1 =1.54051 **Å** for the lower angles and λ2 =1.54433 **Å** for the higher angle.

$s^2 = sum\ of\ square\ of\ miller\ indices\ h^2 + k^2 + l^2$

In a cubic structure, $\frac{sin^2\theta_n}{s_n} = \frac{\lambda^2}{4a^2}$ = constant, with nth row and a= lattice parameter (structure factor)

To calculate the lattice parameter with integer s,

$$a = \left(\frac{\lambda^2}{4a_j^2}\right)^{1/2} \times \frac{\lambda}{2}$$

with jth row

Using Excel, the value obtained for the lattice parameter is roughly 5. 43 for each row.

d_{hkl} can be determined using the formula below. Determine d_{hkl} separately for λ_1 and λ_2 , respectively.

$$2d_{hkl}\sin\theta = \lambda$$

The extrapolation function is obtained by the equation

a= $a_0\, C\, \frac{cos^2\theta}{sin\theta} + a_0$, and the function is $\frac{cos^2\theta}{sin\theta}$

The values a_j are plotted against this function, and a_0 is obtained from the interception of the plot. The extrapolated function is below (Figure 10)

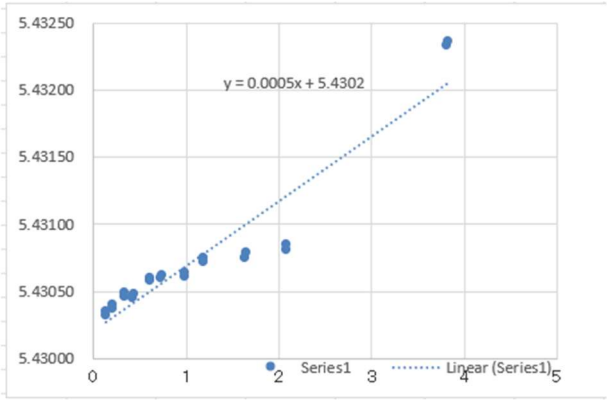


Figure 10 Extrapolation function (Si)

From the extrapolation function, the observed lattice parameter is y-intercept a_0 = 5.4302 **Å** . This is close to the value of the reported value for lattice parameter of Si in the textbook, a= 5.4282 **Å** with very small error (0.03%). Thus, the structure is confirmed to be silicon.

We repeated the same procedures for 2 unknown samples.

ii) Unknown sample 1 (later determined to be Copper, Cu)

The same steps are done for analysis, this time with a different sample. Thus, this time for analysis in Excel wavelengths used are 1.54051 **Å** for lower angles and 1.54433 **Å** for higher angles.

Peak detection of unknown sample 1 (Cu) by High Score software

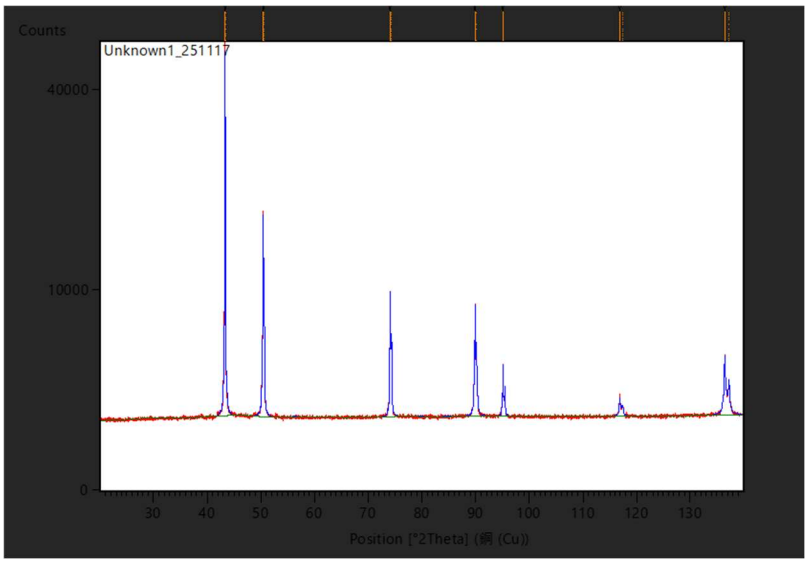


Figure 11 Peak detection of Cu

The diffraction data from High Score is then copied and extracted into Excel (Table 3). This includes the degrees of position which would be used for analysis.

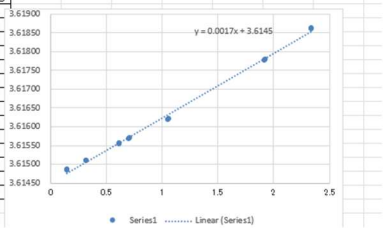
Table 3 Data extracted from High Score

No.	Pos. [° 2	FWHM Lef	計算面積	指定	h	k	l	多重度	F 観測値	F 計算値	Area[cts*° 2	由来	バックグラウンド[cts]								
1	43.2686	0.1239	7015.71						0	0	7015.71	理論KA1	1356.9								
2	43.3816	0.1239	7015.71						0	0	3507.86	理論KA2	1356.9								
3	50.4046	0.1478	3370.59						0	0	3370.59	理論KA1	1325.34								
4	50.5386	0.1478	3370.59						0	0	1685.3	理論KA2	1325.34								
5	74.092	0.1487	1787.03						0	0	1787.03	理論KA1	1328.04								
6	74.3071	0.1487	1787.03						0	0	893.52	理論KA2	1328.04								
7	89.9085	0.1771	1954.72						0	0	1954.72	理論KA1	1325.83								
8	90.1932	0.1771	1954.72						0	0	977.36	理論KA2	1325.83	d値 [Å]	NET強度	K-A2/K-A	一致	一致条件	相対強度	データ元	ステータス	積分幅 [° 2Th.]
9	95.1211	0.1452	576.16						0	0	576.16	理論KA1	1336.9	2.08933	43128.2	0.5	FALSE		100	プロファイルフ	使用	0.16267
10	95.4329	0.1452	576.16						0	0	288.08	理論KA2	1336.9	2.08416	21564.1	0.5	FALSE		50	プロファイルフ	使用	0.16267
11	116.917	0.2926	339.48						0	0	339.48	理論KA1	1343.21	1.809	15950.5	0.5	FALSE		36.98	プロファイルフ	使用	0.21132
12	117.383	0.2926	339.48						0	0	169.74	理論KA2	1343.21	1.80451	7975.22	0.5	FALSE		18.49	プロファイルフ	使用	0.21132
13	136.496	0.314	1564.48						0	0	1564.48	理論KA1	1397.73	1.2786	8142.71	0.5	FALSE		18.88	プロファイルフ	使用	0.21946
14	137.215	0.314	1564.48						0	0	782.24	理論KA2	1397.73	1.27543	4071.35	0.5	FALSE		9.44	プロファイルフ	使用	0.21946
														1.09024	6944.83	0.5	FALSE		16.1	プロファイルフ	使用	0.28146
														1.08754	3472.42	0.5	FALSE		8.05	プロファイルフ	使用	0.28146
														1.04378	2502.98	0.5	FALSE		5.8	プロファイルフ	使用	0.23019
														1.04119	1251.49	0.5	FALSE		2.9	プロファイルフ	使用	0.23019
														0.90383	735.54	0.5	FALSE		1.71	プロファイルフ	使用	0.46154
														0.90159	367.77	0.5	FALSE		0.85	プロファイルフ	使用	0.46154
														0.82935	2971.24	0.5	FALSE		6.89	プロファイルフ	使用	0.52654
														0.8273	1485.62	0.5	FALSE		3.44	プロファイルフ	使用	0.52654

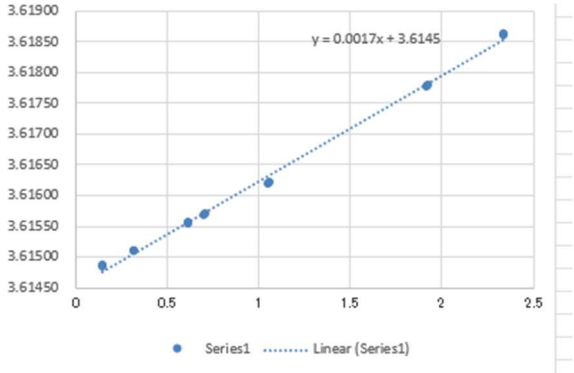
As with the previous Si, the first column from Table 1 is used for the 2θ row in a new table in Excel which is used for the analysis of diffraction and determination of the lattice parameter a, which is specific to each element (Table 4). s, a, hkl and d_{hkl} can be calculated.

Table 4 Unknown sample(Cu) analysis table in Excel with extrapolation function

λ(Å)	2θ	θ	radian	$\sin^2\theta$	$\frac{\sin^2\theta}{\sin^2\theta_1}$	trialS1	$\frac{\sin^2\theta}{\sin^2\theta_1}s_1$	Integer s	h k l	$\frac{\lambda^2}{4a^2}$	a(Å)	Extrapolate	d_{hkl}(Å)	observed	reported	error(%)
1.54051	43.2686	21.6343	0.37759	0.13593	1	3	3	3	111	0.04531	3.61863	2.34369	2.0892	3.6145	3.6153	0.02213
1.54433	43.3816	21.6908	0.37858	0.1366	1.00498		3.01493331	3	111	0.04553	3.61861	2.33605	2.0892			
1.54051	50.4046	25.2023	0.43986	0.18132	1.33396		4.00186726	4	200	0.04533	3.61779	1.92262	1.8089			
1.54433	50.5386	25.2693	0.44103	0.18222	1.34059		4.021774	4	200	0.04556	3.61777	1.91574	1.8089			
1.54051	74.092	37.046	0.64657	0.36295	2.67023		8.010695516	8	202	0.04537	3.61621	1.05742	1.2785			
1.54433	74.3071	37.1536	0.64845	0.36476	2.68352		8.05055904	8	202	0.04559	3.61620	1.0518	1.2785			
1.54051	89.9085	44.9543	0.7846	0.4962	3.6726		11.0178133	11	311	0.04538	3.61570	0.7088	1.0902			
1.54433	90.1932	45.0966	0.78708	0.50169	3.69088		11.0726478	11	311	0.04561	3.61568	0.70354	1.0902			
1.54051	95.1211	47.5606	0.83009	0.54463	4.00682		12.0204718	12	222	0.04539	3.61555	0.61704	1.0437			
1.54433	95.4329	47.7165	0.83281	0.54734	4.02676		12.0802713	12	222	0.04561	3.61553	0.61185	1.0437			
1.54051	116.917	58.4587	1.0203	0.72635	5.34375		16.0312396	16	400	0.0454	3.61510	0.32108	0.9038			
1.54433	117.383	58.6914	1.02436	0.72997	5.37033		16.1110008	16	400	0.04562	3.61509	0.31606	0.9038			
1.54051	136.496	68.2481	1.19115	0.86266	6.34659		19.0397557	19	331	0.0454	3.61485	0.14786	0.8293			
1.54433	137.215	68.6077	1.19743	0.86696	6.37816		19.134481	19	331	0.04563	3.61483	0.14289	0.8293			



The extrapolation function is determined below using the same equations as for Si(Figure 12)



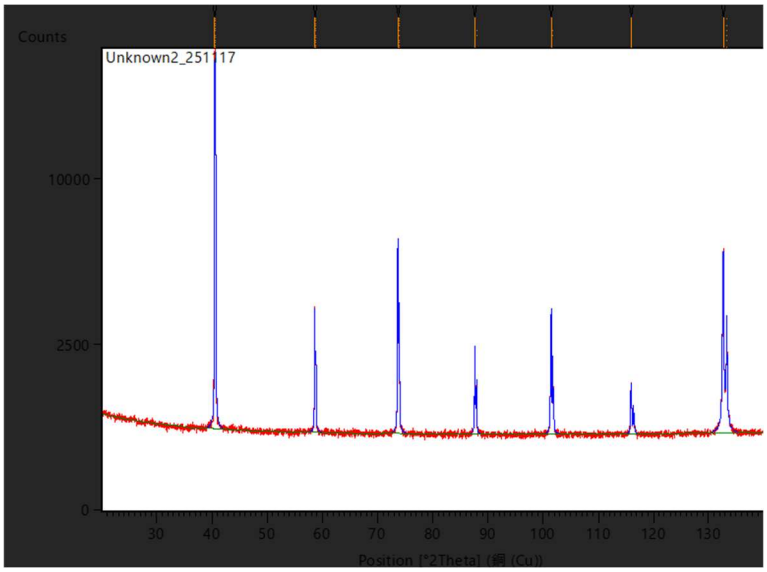
12 Extrapolation function of unknown sample(Cu)

From the extrapolation function, the observed lattice parameter is y-intercept a_0 = 3.6145**Å** . Of all the lattice parameters listed in the textbook, the value is the closest to the reported lattice parameter value of copper, a= 5.4282 **Å** with very small error (0.02213%).

Thus, the unknown sample is determined to be copper (Cu).

iii) Unknown sample 2 (later determined to be Molybdenum, Mo)

X-ray diffraction pattern and peak detection of Mo



13 Peak detection and x-ray diffraction pattern of nknown sample 2(Mo)

The diffraction data from High Score is then copied and extracted into Excel (Table 5). This includes the degrees of position which would be used for analysis in another Excel table (Table 6).

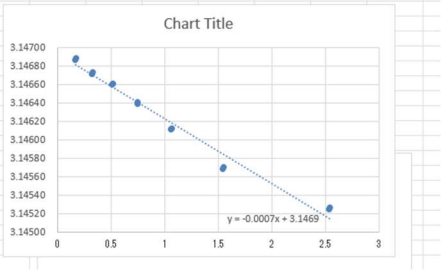
Table 5 Data extracted from High Score

No.	Pos. [° 2Th]	FWHM Left	計算面積	指定	h	k	l	多重度	F 観測値	F 計算値	Area[cts*°	由来	バックグラウンド	d値 [Å]	NET強度	K-A2/K-A1	一致	一致条件	相対強度	データ元	ステータス	積分幅 [° 2Th]
1	40.5262	0.1321	2729.42						0	0	2729.42	理論KA1	601.63	2.22417	16146.2	0.5	FALSE		100	プロファイルフィ	使用	0.16904
2	40.6314	0.1321	2729.42						0	0	1364.71	理論KA2	601.63	2.21865	8073.11	0.5	FALSE		50	プロファイルフィ	使用	0.16904
3	58.6442	0.1241	508.81						0	0	508.81	理論KA1	544.57	1.57294	3059.59	0.5	FALSE		18.95	プロファイルフィ	使用	0.1663
4	58.8042	0.1241	508.81						0	0	254.41	理論KA2	544.57	1.56904	1529.8	0.5	FALSE		9.47	プロファイルフィ	使用	0.1663
5	73.6968	0.1271	1080.98						0	0	1080.98	理論KA1	532.52	1.28447	6074.82	0.5	FALSE		37.62	プロファイルフィ	使用	0.17794
6	73.9103	0.1271	1080.98						0	0	540.49	理論KA2	532.52	1.28129	3037.41	0.5	FALSE		18.81	プロファイルフィ	使用	0.17794
7	87.6426	0.1298	366.26						0	0	366.26	理論KA1	519.04	1.11249	1918.58	0.5	FALSE		11.88	プロファイルフィ	使用	0.1908
8	87.9161	0.1298	366.26						0	0	183.13	理論KA2	519.04	1.10973	959.29	0.5	FALSE		5.94	プロファイルフィ	使用	0.1908
9	101.445	0.1316	634.38						0	0	634.38	理論KA1	520.82	0.9951	3132.62	0.5	FALSE		19.4	プロファイルフィ	使用	0.20251
10	101.794	0.1316	634.38						0	0	317.19	理論KA2	520.82	0.99264	1566.31	0.5	FALSE		9.7	プロファイルフィ	使用	0.20251
11	115.977	0.1588	231.69						0	0	231.69	理論KA1	519.79	0.90843	921.42	0.5	FALSE		5.71	プロファイルフィ	使用	0.25145
12	116.434	0.1588	231.69						0	0	115.84	理論KA2	519.79	0.90618	460.71	0.5	FALSE		2.85	プロファイルフィ	使用	0.25145
13	132.649	0.1959	1773.36						0	0	1773.36	理論KA1	551.33	0.84109	5538.03	0.5	FALSE		34.3	プロファイルフィ	使用	0.32022
14	133.303	0.1959	1773.36						0	0	886.68	理論KA2	551.33	0.839	2769.01	0.5	FALSE		17.15	プロファイルフィ	使用	0.32022

The first column from Table 5 is used for the 2θ row in a new table in Excel which is used for the analysis of diffraction and determination of the lattice parameter a, which is specific to each element (Table 6). The wavelengths used are also 1.54051 **Å** for lower angles and 1.54433 **Å** for higher angles. s, a, hkl and d_{hkl} can be calculated.

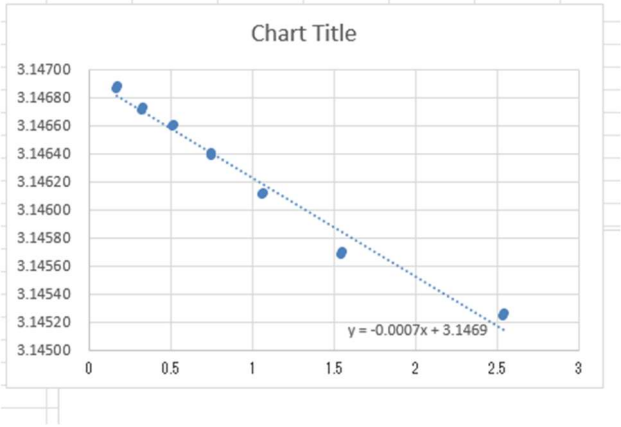
Table 6 Analysis table in Excel for Unknown Sample 2(Mo)

λ(Å)	2θ	θ	radian	$\sin^2\theta$	$\frac{\sin^2\theta}{\sin^2\theta_1}$	trialS1	$\frac{\sin^2\theta}{\sin^2\theta_1}$	Integer s	h k l	$\frac{\lambda^2}{4a^2}$	a(Å)	Extrapolate	d_{hkl}(Å)	observed	reported	error(%)
1.54051	40.5262	20.2631	0.35366	0.11995	1	2	2	2	110	0.05997	3.14527	2.54108	2.2240	3.1469	3.1466	0.00953
1.54433	40.6314	20.3157	0.35458	0.12054	1.00498		2.00996	2	110	0.06027	3.14525	2.53305	2.2240			
1.54051	58.6442	29.3221	0.51177	0.23982	1.99944		3.99889	4	200	0.05996	3.14570	1.55227	1.5729			
1.54433	58.8042	29.4021	0.51316	0.24102	2.00939		4.01879	4	200	0.06025	3.14569	1.54599	1.5728			
1.54051	73.6968	36.8484	0.64313	0.35964	2.99836		5.99672	6	211	0.05994	3.14613	1.0678	1.2844			
1.54433	73.9103	36.9552	0.64499	0.36143	3.01328		6.02655	6	211	0.06024	3.14611	1.06218	1.2844			
1.54051	87.6426	43.8213	0.76483	0.47943	3.99709		7.99419	8	202	0.05993	3.14641	0.75182	1.1124			
1.54433	87.9161	43.9581	0.76721	0.48182	4.01698		8.03395	8	202	0.06023	3.14640	0.74652	1.1124			
1.54051	101.445	50.7225	0.88527	0.59921	4.99571		9.99143	10	310	0.05992	3.14662	0.51775	0.9950			
1.54433	101.794	50.8969	0.88832	0.6022	5.02057		10.0411	10	310	0.06022	3.14660	0.51263	0.9950			
1.54051	115.977	57.9884	1.01209	0.719	5.99442		11.9888	12	222	0.05992	3.14673	0.33139	0.9084			
1.54433	116.434	58.2169	1.01608	0.72258	6.02424		12.0485	12	222	0.06022	3.14672	0.32636	0.9084			
1.54051	132.649	66.3246	1.15758	0.83875	6.99279		13.9856	14	321	0.05991	3.14689	0.17606	0.8410			
1.54433	133.303	66.6514	1.16329	0.84293	7.02758		14.0552	14	321	0.06021	3.14687	0.17108	0.8410			



The same steps are done for analysis, this time with a different sample.

The extrapolation function is shown in the graph below



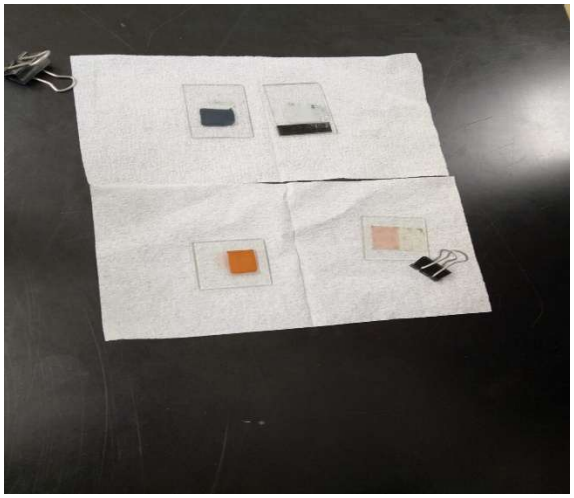
14 Extrapolation function

Thus, for analysis:

From the extrapolation function, the observed lattice parameter is y-intercept a_0 = 3.1469**Å** . Of all the lattice parameters listed in the textbook, the value is the closest to the reported lattice parameter value of molybdenum, a= 3.1466 **Å** with very small error (0.00953%). Of all the three samples examined, this is the smallest error.

Thus, the unknown sample is determined to be molybdenum (Mo).

After the experiment, the samples of the unknown samples (right side of Figure 15) are compared to Cu (orange-coloured) and Mo (black-coloured) thin plate prototypes (left side of Figure 15). Thus, by observation it is confirmed that the samples are Cu and Mo, respectively.



15 Comparison of Cu and Mo samples to prototypes

3) Structure factor

The structural type of each sample is in the table below

Sample	Structure type
LiF	NaCl-type
Si	Diamond type
Cu	Face-centered cubic
Mo	Body-centered cubic

a) Structure factor of Li

Li has a rock salt (NaCl) structure.

The structure factor for:

h,k,l unmixed:

$$F_{hkl} = 4f_{Na} + f_{Cl}e^{-\pi i(h+k+l)}$$
$$\begin{cases} = f_{Na} + f_{Cl} & ; hkl \text{ all even,} \\ = f_{Na} - f_{Cl} & ; hkl \text{ all odd.} \end{cases}$$

and h,k,l mixed: $F_{h,k,l}$=0

Therefore, the indices 001, 110, 210, 211 vanish. (Systematic absence)

b) Structure factor of Si:

Si has a diamond structure.

The structure factor for:

h,k,l unmixed:

$$F_{hkl} = 4f1 + e^{-\pi i(h+k+l)/2}$$
$$\begin{cases} = 8f & ; h+k+l = 4m, \\ = 4f(1 \mp i) & ; h+k+l = 4m \pm 1 \\ = 0 & ; h+k+l = 4m \pm 2 \end{cases}$$

and h,k,l mixed: $F_{h,k,l}$=0

Therefore, the indices 001, 110, 210, 211 vanish.

From Table 2 of Si (Diffractometer experiment), it can be seen that there are no mixed values (odd-even) for h,k,l indices.

c) Structure of Cu:

Cu has a face-centered cubic structure. The structure factor is calculated below:

$$F_{hkl} = f\{1 + e^{-\pi i(k+l)} + e^{-\pi i(h+l)} + e^{-\pi i(h+k)}\}$$
$$\begin{cases} = 4f & ; h,k,l \text{unmixed} \\ = 0 & ; h,k,l \text{mixed} \end{cases}$$

From Table 4 of Cu(Diffractometer experiment), it can be seen that there are also no mixed values(odd-even) for h,k, l indices. This is suitable according to the structure factor of Cu.

d) Structure of Mo:

Mo has a body-centered cubic structure. The structure factor is calculated below:

$$F_{hkl} = f\{1 + e^{-\pi i(h+k+l)}\}$$
$$\begin{cases} = 2f & ; h+k+l \text{ is even}, \\ = 0 & ; h+k+l \text{ is odd}. \end{cases}$$

From Table 6 of Mo(Diffractometer experiment), there are no h,k,l indices that add up to an odd number. This is suitable according to the structure factor of Mo.

From the diffractometer experiment method, there are several ways to estimate and confirm the element (name) of an unknown sample crystal:

1) Calculating the lattice parameter and creating an extrapolation function to find the y-intercept a0, and then compare this to the reported value, see which element has the closest lattice parameter
2) Calculate h, k, l indices and calculate the structure factor

E. Discussion

a) Laue method:

1) How systematic absence is reflected in the Laue spots (Structure factor $F_{h,k,l}$=0):

If spots are missing on the Laue picture when compared to a normal projection, those are the spots where systemic absences happen. By comparing the Laue picture with a normal projection, we can see which spots are not there. We then use tools like a stereographic ruler and the standard projection to label those spots and find out which Miller indices are causing systemic absences.

Systematic absences in Laue patterns are used to identify space group symmetry and determine allowed reflections.

2) The symmetry observed in the Laue pattern

Laue photos of each three samples are taken three times, each time with different cut of the crystal, respectively, and the better images are chosen for further analysis.

For LiF(001), the symmetry was slightly skewed but almost symmetrical, showing a 4-fold symmetry. Si(001) was heart shaped, not a lot of symmetry and slightly skewed. However, the image of Si(111) is asymmetrical and can only be approximated by a three-fold symmetry. The reason for the skewing might be because the sample is not placed exactly perpendicular in the X-ray machine, or/and of the quality of the sample itself. Each group member uses

different cuts/geometries of samples, which will also affect the quality and the symmetry.

b) Diffractometer method

-Difference between measured and literature values of lattice constants

Sample	Measured(diffractometer)	Literature	Difference(error)
Si	5.4302	5.4282	0.036831%
Cu	3.6145	3.6153	0.022133
Mo	3.1469	3.1466	0.009533

The difference, though very small, might be due to:

- Measurement error: arising from a constant shift in peak position, and errors arising from a Bragg angle dependent shift in peak position.
- Error and imprecision in extrapolation of data values gained from analysis of powdered x-ray diffraction of samples
- Multiplying a multiplication factor to the determined parameter(s)
- Round-off or cut off decimals in Excel to solve equations to determine values of parameters
- Inconsistent intensity of x-ray due to voltage fluctuation or impurities within the crystal.

- Summary of identifying substances of “unknown” samples:

1) Obtain diffraction pattern of the sample and the peaks using the X-ray specification
2) Match diffraction peaks to their (hkl) miller indices.

3) Find the angle of each peak using the dhkl spacing.
4) Compute for each peak $sin^2\theta$
5) Extract the lattice constant a
6) Refine the lattice constant using the extrapolation method
7) Match the lattice constant with literature(find the closest lattice parameter in literature) to identify the substance.